A GraphCast-style fitting model as Plexus2 operators

Stages 0–1b: the container, the two-scale toy, and the gates it must pass

`Plexus/prototype/graphcast` — prototype only, nothing promoted

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Abstract

This prototype expresses a GraphCast-style dynamics model as a composition of Plexus2 sets, fields and operators, so that one spec language covers a toy with ground truth, a point-cloud recording and a field recording. Four architectural choices — an encoder/decoder through a background grid, the message form, the number of message-passing passes, and the node embedding — are *configuration options* rather than forks, giving 24 legal combinations from one implementation. This note covers stages 0 to 1b: the container, the two-scale toy, and the nine gates now passing. **No model has been gated yet**, and that is deliberate: three successive toys failed for reasons that had nothing to do with any model, and stage 1b exists because in each case training had been run before the data was known to pose the problem it claimed to.

1. Mechanism

Every model in `connectome-gnn` is a one-step message-passing network on a *given* connectome. Two things wanted now sit outside that: a deeper, edge-stateful processor, and datasets with no connectome at all. The zebrafish recording has 71,721 neurons of which about 481 are matched to an electron-microscopy reconstruction; the organoid is a three-dimensional field with no segmentation. In both, the coupling has to come from *space* rather than from wiring.

The arithmetic that decides what can be represented. A zebrafish volume is acquired plane by plane — 72 planes at 12.0 ms, so one volume takes 0.851 s against a 0.914 s frame interval, which is 93% of the frame period. Neurons at different depths are therefore *not* sampled simultaneously, and simultaneity is exactly what a message-passing operator assumes. That is a fact about the data rather than a modelling choice; the per-plane acquisition times are recorded, so the model can be conditioned on the offset rather than pretending it away.

The toy is two scales with deliberately different rules. A test bed whose coarse and fine mechanisms are the same rule at two resolutions cannot show whether a multi-scale operator is doing anything at all. So:

scale	operator	rule
coarse	`wave_field`	a wave travelling cyclically left to right; one rule, no per-node freedom
fine	`gradient_gain`	each node amplifies the *local gradient* of that field with its own *signed* gain

The heterogeneity lands in the fine rule, as a signed per-node gain: some nodes amplify the gradient, others invert it. Unlike a time constant, that cannot be read off a node's own trace without reference to the field around it — which is what makes the graph load-bearing rather than a small correction to a mostly-local law.

2. Equations and symbols

$$\textbf{coarse:}\quad \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0, \qquad u(x,t) = A\sin\Big(2\pi\Big(\frac{x}{\lambda} - \frac{t}{T}\Big)\Big), \qquad c = \frac{\lambda}{T} \tag{1}$$

$$\textbf{fine:}\quad \frac{\mathrm{d}v_i}{\mathrm{d}t} = -\frac{v_i}{\tau_i} \;+\; g_i\,\frac{\partial u}{\partial x}(\mathbf{r}_i) \tag{2}$$

The coarse field is written in closed form rather than stepped, and that is deliberate: it is the exact solution of (1), so the coarse scale carries no discretisation error of its own that could later be mistaken for an error in the rule under test.

symbol	units	what it is *of*
$u(x,t)$	field units	the coarse field, one scalar per point of the domain
A	field units	wave amplitude
λ	length	wavelength, *as a fraction of the domain width*
T	frames	wave period, in simulation frames
$c = \lambda/T$	length / frame	phase speed — the quantity gate G21 measures
v_i	state units	the state of node i; the observable
τ_i	time	node i's time constant, a per-*type* constant
g_i	state / (field / length)	node i's **signed gain** on the field gradient — the heterogeneity
$\mathbf{r}_i$	length	node i's position, fixed

τ_i and g_i are intensive: properties of a node rather than sums over it, so neither changes if the node set is resampled at a different density. That matters because the same operators are meant to run on 1,024 toy nodes and on 71,721 real ones.

3. The Plexus2 container

operator	kind	acts on	mechanism
`wave_field`	`field`	field `u`	Eq. (1); prescribed, exact
`gradient_gain`	`exchange`	`neuron` $\leftarrow$ `u`	Eq. (2); `EMIT = velocity`, so the engine integrates it

Both are prototype-local, registered through the ordinary `@register_operator` decorator; nothing is written into `src/plexus/`. The forward run is `plexus.engine.run` unchanged, and the interaction graph is built from the node positions by the *fit*, never by the generator — building the generator on the same graph the model is given would hand it the answer.

The four options. `encoder_decoder` (`off` / `on`), `message` (`simple` / `graphcast`), `n_passes` (≥ 1), and `embedding` (`none` / `free` / `multires`). The schema refuses `simple` at more than one pass — that form carries no edge state, so repeating it is a different model from the one it claims to be — which leaves $2 \times 4 \times 3 = 24$ legal combinations, the count G1 checks.

Where the training scheme comes from. The optimiser follows GraphCast where its choices are measured (AdamW with $\beta_2 = 0.95$, a global gradient-norm clip, a linear warmup then cosine decay *to zero*, and the target normalised by the inverse variance of the *increment* rather than of the state), and `connectome-gnn`'s production configs elsewhere: a separate learning rate for the interaction weights, L_1 and L_2 on those weights, a Lipschitz smoothness prior on the message function, and a group lasso over the message's input blocks. The last was measured in the weekend benchmark to recover +0.153 in R^2_W on five of five folds by removing a degeneracy rather than by preferring simpler models — which is why accuracy rose rather than trading off. The rollout knobs the same benchmark separated (horizon schedule, truncation window, shooting stride, step weighting) are implemented so its finding — that plain one-step training wins — can be reproduced here rather than assumed.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. A threshold chosen after seeing the number is not a threshold, so every threshold is a literal in `gates.py` and predates the code it gates. This table is 9 bookkeeping / 13 closed form / 4 measurement, so two thirds of it is verification rather than validation. That is the expected shape and is stated rather than hidden.

A gate's threshold belongs in the unit of the phenomenon, not of the mesh: a threshold in grid cells is denominated in the currency in which it is easiest to pass, and a green row of that kind reads as an endorsement the model has not earned. Every row therefore carries what its number is *of*, and the measurement tier is available only because the spec declares `general.units:`. **An artifact is a condition of passing** — a gate with no figure is reported SKIP, never **PASS**.

id	gate	threshold	measured	outcome	figure
Bookkeeping — does the code do what the operator says?					
G1	all four options PARSE (24 combinations)	24 of 24 option combinations load	24	PASS	G1_options.png
G1b	all four options BUILD and take one step	24 of 24 option combinations run one forward step	—	SKIP	—
G2	nothing dataset-specific is hardcoded	0 offending literals outside config/	0	PASS	G2_scan.png
G3	scatter->gather round trip on a constant field	< 1e-6 of the field value	—	SKIP	—
G4	transfer weights are a partition of unity	|sum(w) - 1| < 1e-6	—	SKIP	—
G5	simple + 1 pass + no enc/dec is arithmetically NeuralGNN	< 1e-5 of the voltage range	—	SKIP	—
G6	residual blocks start at identity: 1 vs 16 passes at init	bit-identical (max |delta| == 0)	—	SKIP	—
G7	units declared, and every measurement threshold is in phenomenon units	1 = declared and checked	1	PASS	G7_units.png
G8	a K=20 rollout on the toy does not diverge	state norm stays < 2x the ground-truth norm	—	SKIP	—
Closed form — does it reproduce the physics it was given?					
G9	the message becomes a gradient operator	R^2 > 0.90 against the true field gradient	—	SKIP	—
G10	recover the per-node time constant	R^2 > 0.95 against the known tau	—	SKIP	—
G11	the embedding recovers the types (small toy)	ARI > 0.70 against the 6 true types	—	SKIP	—
G12	the embedding recovers the types (large toy)	ARI > 0.70 against the 65 true types	—	SKIP	—
G13	recover the per-node SIGNED GAIN (the heterogeneity)	R^2 > 0.90 against the true g_i	—	SKIP	—
G14	encoder/decoder is a genuine option: on vs off	|delta R^2(gradient)| < 0.03	—	SKIP	—
G15	graphcast vs simple message is RESOLVED, either way	|delta| reported against the 3-seed floor; below it is UNRESOLVED, not ranked	—	SKIP	—
G16	types are spatially mixed by construction	spatial-cell type purity within 20% of chance (1/n_types)	1.131	PASS	G16_toy.png, G16_state.mp4
G21	the coarse field is a travelling wave, cyclic left to right	phase drift per frame within 5% of lambda/period	0.005224	PASS	G21_field.mp4
G22	the fine rule is exactly recoverable from (v, grad u)	minimum per-node R^2 > 0.90	0.9981	PASS	G22_identifiability.png
G23	the gradient is reconstructible from neighbours' states	R^2 > 0.95, else the graph cannot carry the fine rule	1	PASS	G22_identifiability.png
G24	the heterogeneity is linearly readable	corr(fitted gain, true g_i) > 0.90	0.9807	PASS	G24_heterogeneity.png
G25	connected nodes are not collinear	mean |corr| between connected nodes < 0.80	0.6074	PASS	G22_identifiability.png
Measurement — does it agree with something observed?					
G17	ZAPBench held-out prediction of d(dF/F)/dt	R^2 > 0.268, the parameter-free kNN spatial pool	—	SKIP	—
G18	the learned stimulus gain b_i is spatially structured	Moran's I > 0.2 over the soma graph, against a permutation null	—	SKIP	—
G19	fitted calcium decay time constant	0.5 - 2 s (GCaMP6)	—	SKIP	—
G20	redox field fit reproduces the washout response	THRESHOLD TO BE FIXED from Develop-	—	SKIP	—

4.1 What the gates license, and what they do not

Stages 0 and 1b have run. G1, G2 and G7 establish that all 24 option combinations parse, that no dataset path, name or dimension appears anywhere in the prototype's code, and that the spec declares its units — so the measurement-tier rows are *available* to be denominated in seconds and micrometres rather than in cells. G16 and G21–G25 establish that the toy is a valid test bed: the coarse field really is a travelling wave at the speed asked for, the fine rule is exactly recoverable from the state and the gradient, the gradient is reconstructible from a node's neighbours, the heterogeneity is linearly readable, and connected nodes are not collinear.

Nothing about any model is licensed. Every gate on a fitted model is still SKIP. G5 is the one to watch: it establishes that the `simple` option is arithmetically the existing `NeuralGNN`, and therefore that everything after it is a controlled variation on something already known to work rather than a new model whose failures cannot be attributed.

4.2 Two gates failed first, and both failures were worth having

- **G25 at** 0.841 **was a real defect.** With $\lambda = 0.5$, a twelve-neighbour neighbourhood spans only 0.5 radians of phase, so neighbours were very nearly redundant — which is precisely why an earlier fit drove the loss to 0.005 while recovering none of the mechanism. Shortening to $\lambda = 0.15$ moved it to 0.607.

- **G21 at** 0.094**, then** 0.053**, was the estimator twice over.** Taking the wave peak by `argmax` on a 128-cell grid quantises to whole cells while the wave moves about half a cell per frame; replacing that with an FFT then used integer bin 7 where the true mode is $1/\lambda = 6.67$, biasing the speed by exactly $6.67/7 = 0.952$ — which was the 5% it reported. Projecting onto the known wavelength gives 0.5% error. *An estimator has to be sharper than the threshold it is judged against.*

A third defect never reached a gate because a figure caught it first: nodes were spawned as a disc about a randomly placed parent, so 58.8% of them sat *outside* the field domain, where sampling clamps and the gradient is identically zero. The correlation between a node's fit quality and its distance from the domain edge was 0.863, which is what pointed at it.

5. The evidence

Every gate points to a figure, because the failures this prototype is most exposed to are visible and not scalar. Three defects in earlier toys were each a plausible-looking number: a circuit sitting at a fixed point (spatial spread 9.29 against a temporal spread of 0.065), a stimulus field that was *identically zero* for three generations because an operator read a clock that nothing was writing, and a spatial type purity of 6.1× chance that was the grid being finer than the sampling. All three are obvious in a panel and invisible in a table.

G16 also has a movie: G16_state.mp4.

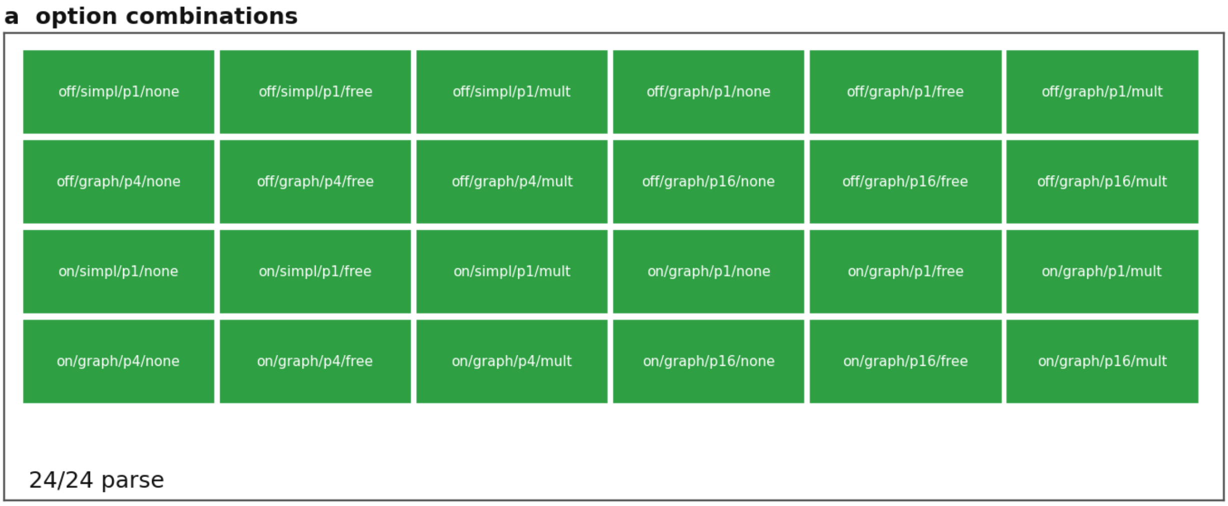

Figure 1: **G1** — all four options PARSE (24 combinations). Threshold: 24 of 24 option combinations load. Measured: 24. Outcome: PASS.

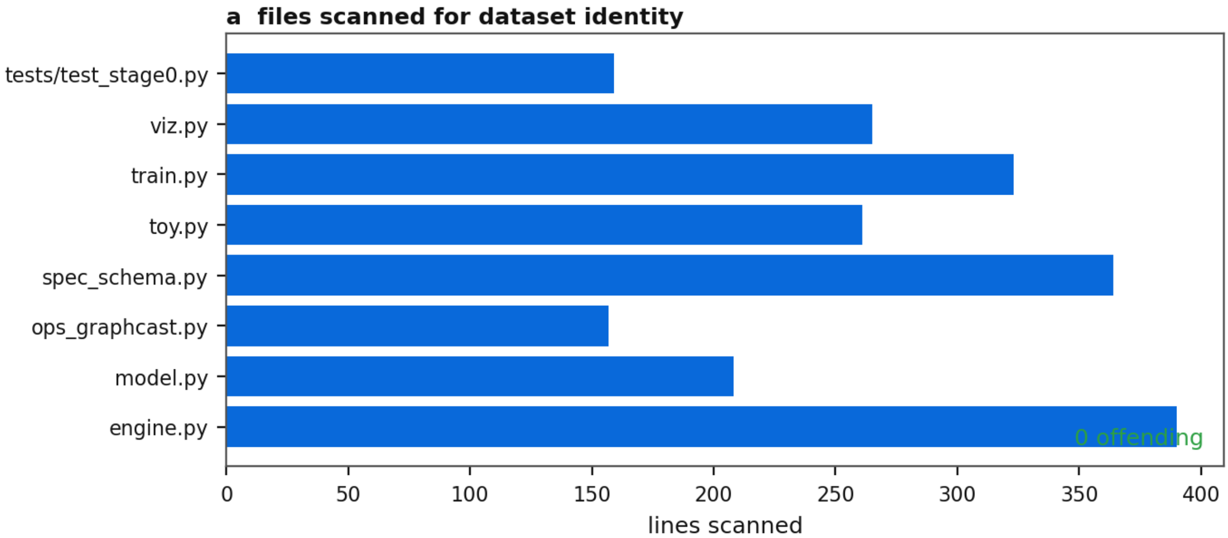

Figure 2: **G2** — nothing dataset-specific is hardcoded. Threshold: 0 offending literals outside config/. Measured: 0. Outcome: PASS.

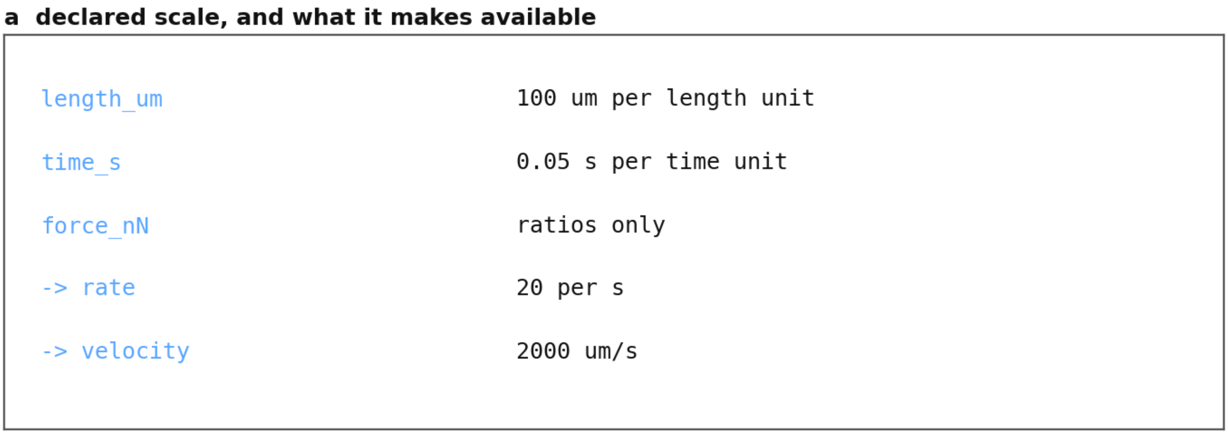

Figure 3: **G7** — units declared, and every measurement threshold is in phenomenon units. Threshold: 1 = declared and checked. Measured: 1. Outcome: PASS.

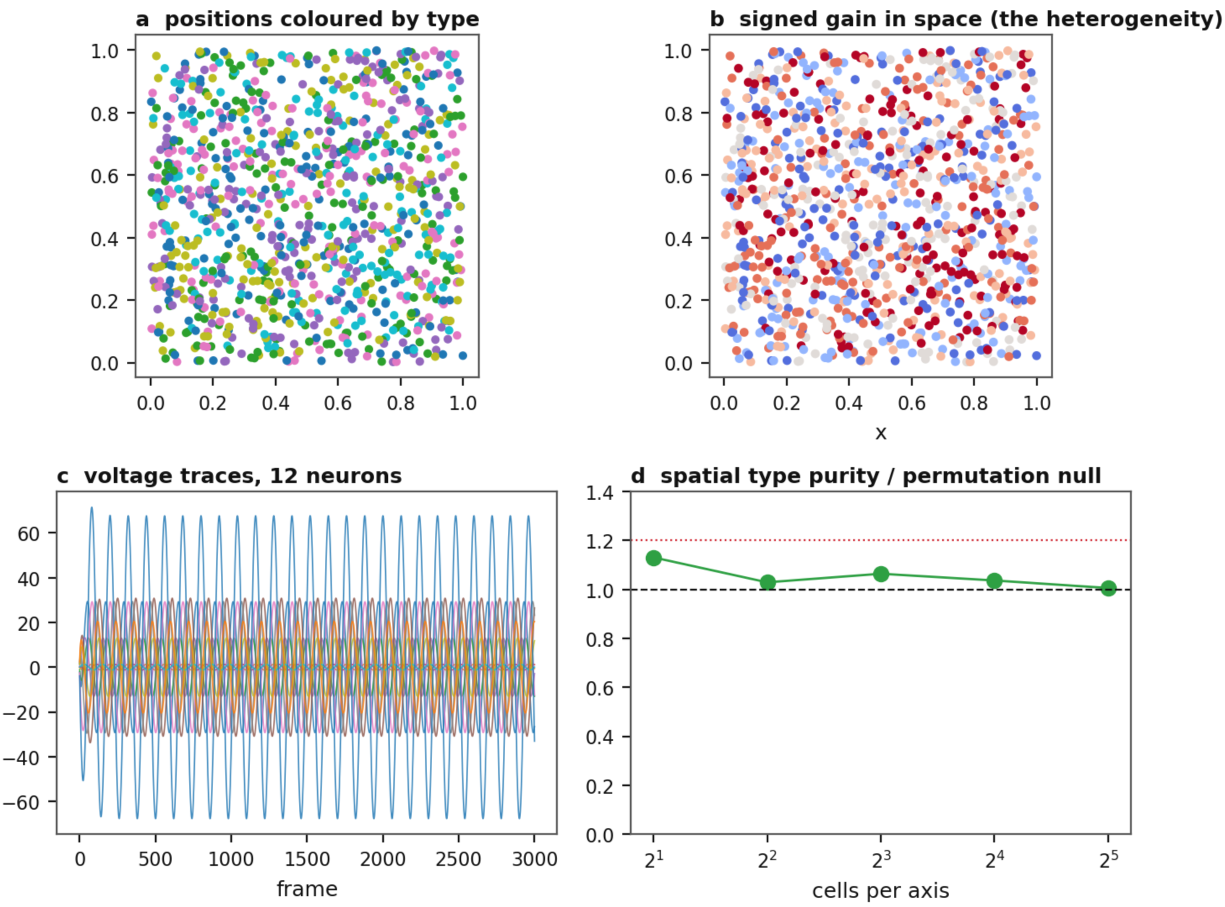

Figure 4: **G16** — types are spatially mixed by construction. Threshold: spatial-cell type purity within 20% of chance (1/n_types). Measured: 1.131. Outcome: PASS.

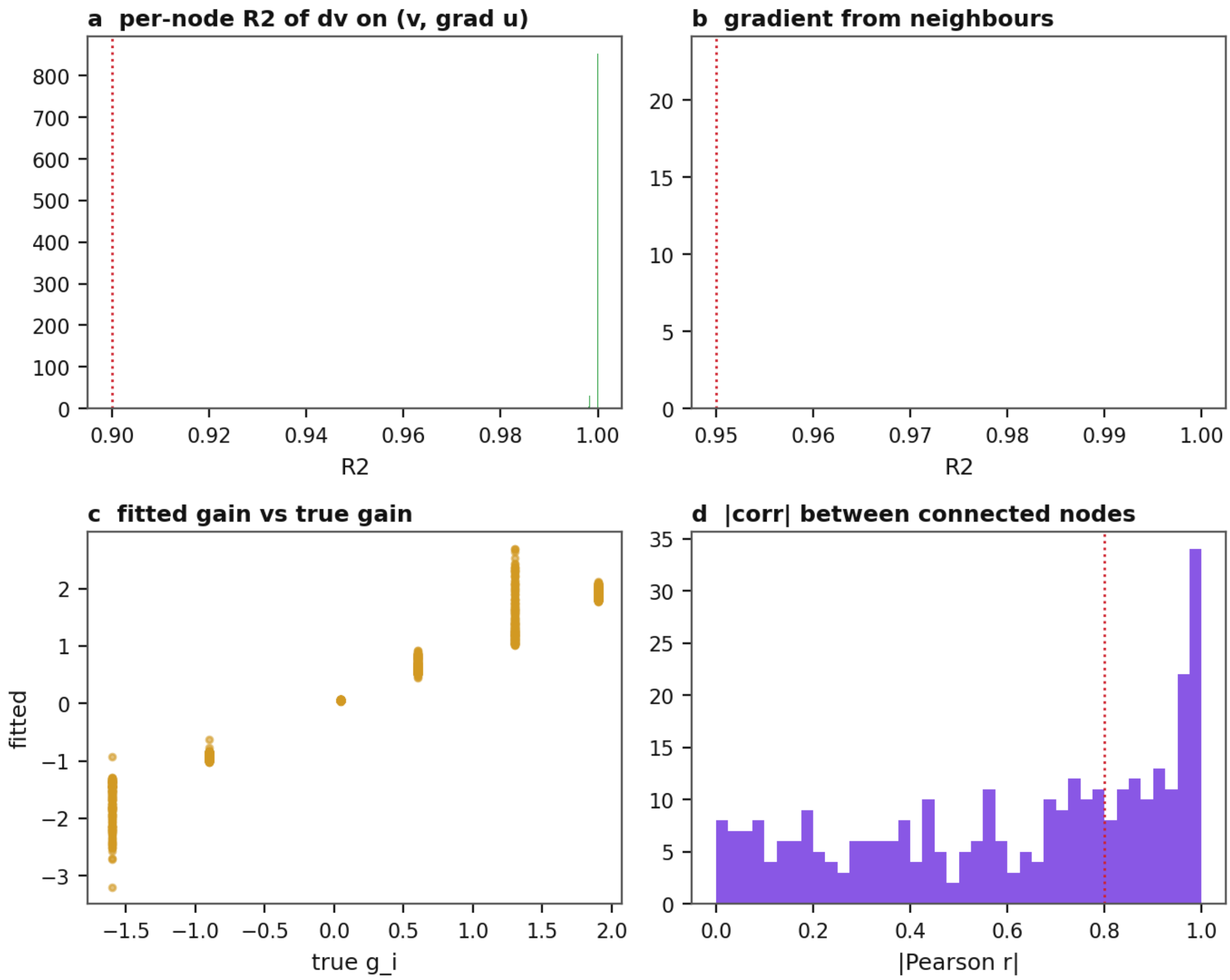

Figure 5: **G22** — the fine rule is exactly recoverable from (v, grad u). Threshold: minimum per-node R^2 > 0.90. Measured: 0.9981. Outcome: PASS.

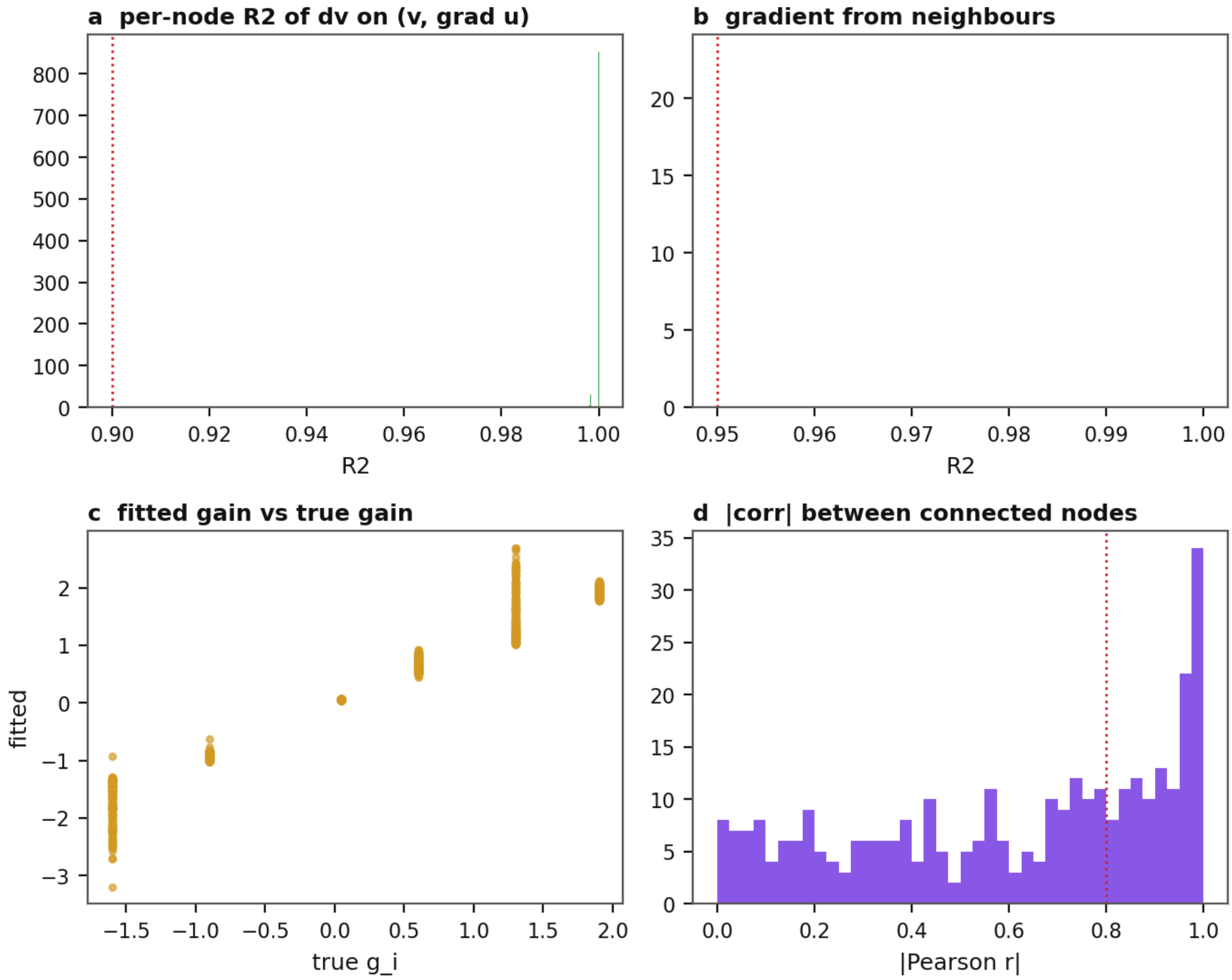

Figure 6: **G23** — the gradient is reconstructible from neighbours' states. Threshold: R^2 > 0.95, else the graph cannot carry the fine rule. Measured: 1. Outcome: PASS.

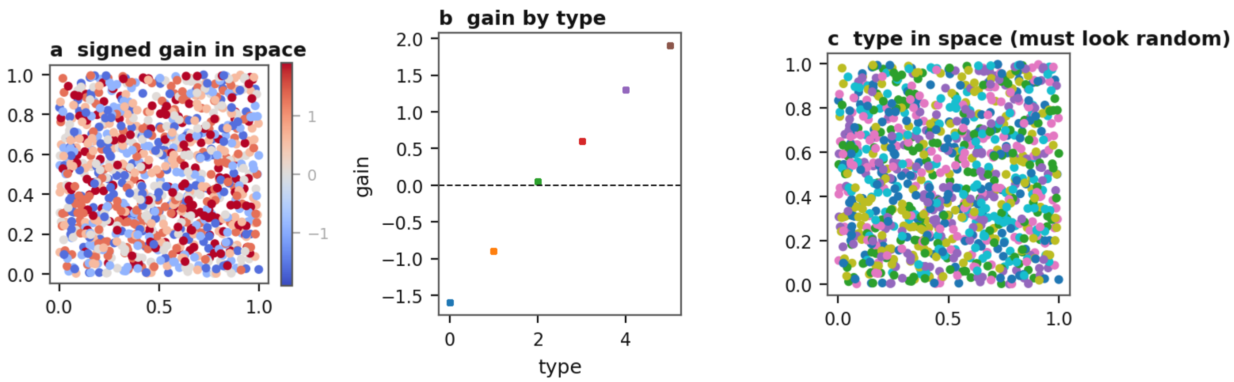

Figure 7: **G24** — the heterogeneity is linearly readable. Threshold: corr(fitted gain, true g_i) > 0.90. Measured: 0.9807. Outcome: PASS.

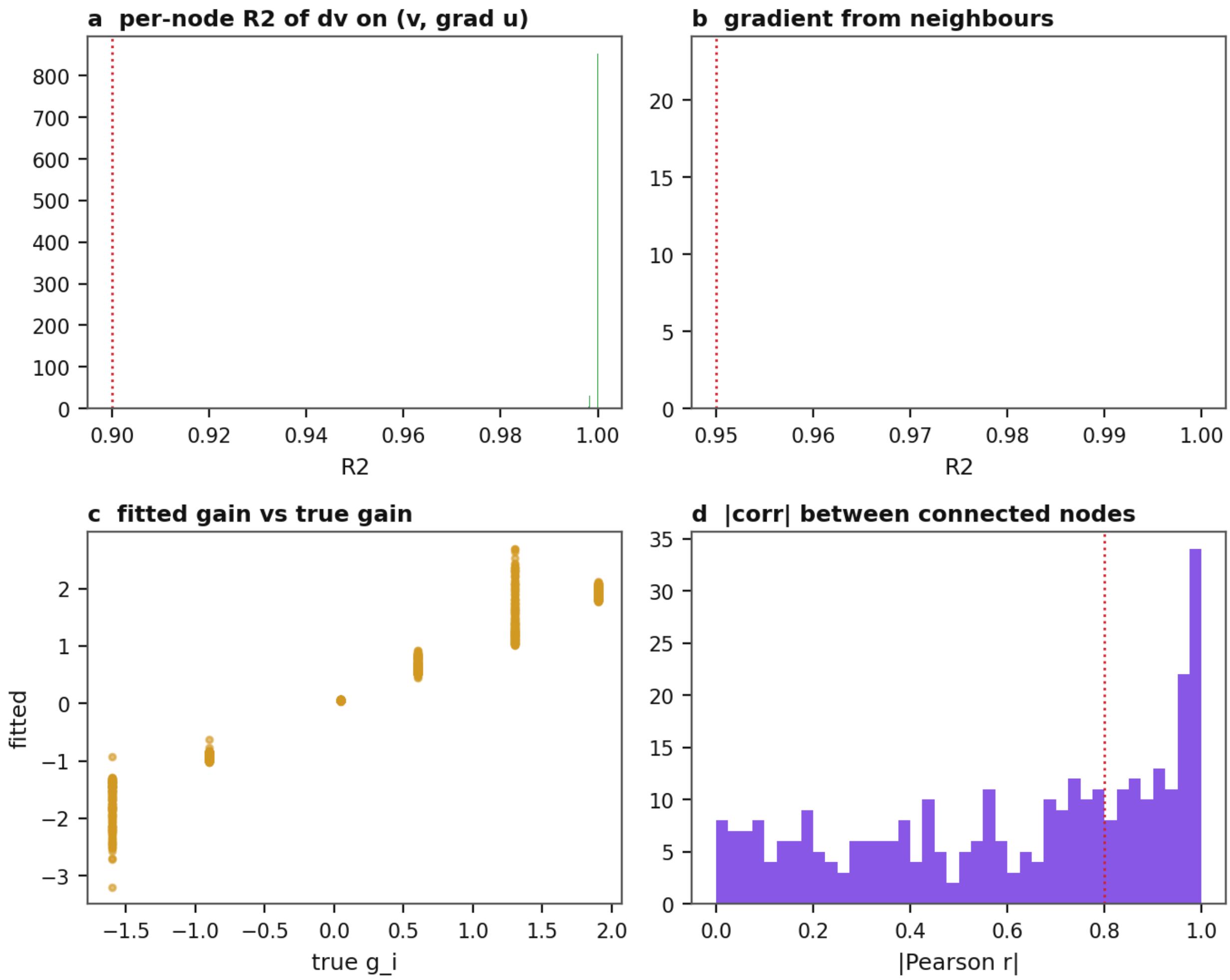

Figure 8: **G25** — connected nodes are not collinear. Threshold: mean |corr| between connected nodes < 0.80. Measured: 0.6074. Outcome: PASS.